

Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without soldering; apply a thin hydrophobic or low-surface-energy coating (e.g., PTFE or silane-based) to the coil exterior and shroud interior to reduce scale adhesion, and add a short straight inlet diffuser to stabilize flow before the helical section.

Executive Technology Brief

Domain: mechanical_structural

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'scale deposition and pitch deformation in helical coil' concerning countercurrent heat recovery module for espresso machine. User-submitted engineering problem (session input): "A compact countercurrent heat recovery module for commercial espresso machines: clean incoming water is preheated by waste heat from the steam boiler exhaust using a coiled helical copper tube wound at uniform pitch inside a cylindrical stainless steel shroud, with flanged inlet and outlet nozzles a". ROUTING: nhtsa_complaints: routing state ARCHIVED_ROUTING — measured: NOT_QUERIED_GRAMMAR on every non-vehicle problem (70 recorded grammar mismatches, zero production-run uses in the current domain set — R399 audit W3); reinstatement: the production domain set includes vehicle problems with extracted make|model|year parameters (flip to ACTIVE + worklog entry; connector implementation archived in place). ROUTING: doe_osti: routing state SUSPENDED_RELEVANCE — measured relevance instability (R375: same fusion papers relevant 3 -> 0; pooled relevant-rate 0.175 pre-fix; DETERMINISM_GAP flagged in the R394 production audit) — records poison synthesis-feeding retrieval more than they feed it until relevance is semantically adjudicated; reinstatement: the Phase P1 semantic reranker lands (semantic relevance adjudication between query and record, replacing lexical overlap); flip to ACTIVE + a measured relevance re-run + worklog entry. [nhtsa_complaints] retrieval status NOT_QUERIED_ARCHIVED_ROUTING (engine routing decision (R399 W3): ARCHIVED_ROUTING — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). [doe_osti] retrieval status NOT_QUERIED_SUSPENDED_RELEVANCE (engine routing decision (R399 W3): SUSPENDED_RELEVANCE — not a provider failure and not absence) — provider outcome recorded, not treated as absence (Art. XXI.3). Problem-existence basis: NO SOURCE RETURNED RECORDS — problem existence UNVERIFIED by live evidence (Art. XX); treat as hypothesis.

Mechanism: The helical coil is treated as a self-supporting structural spring whose pitch stability comes from elastic preload and distributed mechanical constraint, not from metallurgical bonding; scale deposition is reduced by maintaining a continuous thin-film flow regime and by making the coil surface energetically unfavorable to calcium carbonate nucleation.

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Wind the copper tube on a mandrel with a controlled axial preload so the coil is slightly compressed against the cylindrical shroud inner wall, then install a removable slotted pitch-retention cage (stainless steel or PEEK) that engages each coil turn at three or more circumferential points without soldering; apply a thin hydrophobic or low-surface-energy coating (e.g., PTFE or silane-based) to the coil exterior and shroud interior to reduce scale adhesion, and add a short straight inlet diffuser to stabilize flow before the helical section.

Governing Model: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value – geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: APPLICABLE; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation – use their measured plateau relations instead. Engaged variables: epsilon.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: UNKNOWN (pre-registered pass/fail required before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.